

Monday warm-up: Forces IV

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October 6, 2025

1 Memory Bank

- Young's Modulus, Y , has units of N m^{-2} , and it relates the change in length ΔL of a system of original length L_0 and cross-sectional area A subject to a force F :

$$\frac{\Delta L}{L_0} = \frac{1}{Y} \frac{F}{A} \quad (1)$$

Summarized as $\text{stress} = Y \times \text{strain}$.

- Shear Modulus, S , has units of N m^{-2} , and it relates the sideways change in length Δx of a system of length L_0 and cross-sectional area A subject to a force F :

$$\frac{\Delta x}{L_0} = \frac{1}{S} \frac{F}{A} \quad (2)$$

Summarized as $\text{stress} = S \times \text{shear}$.

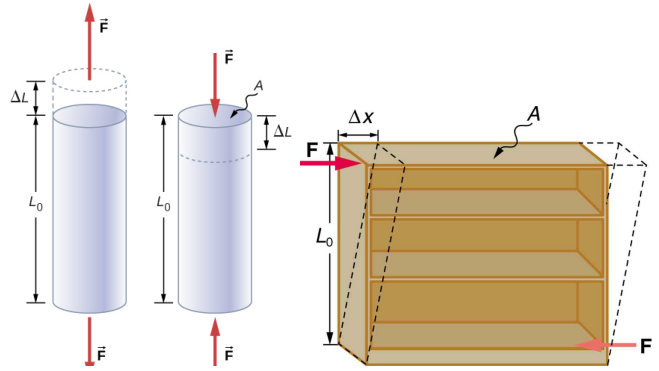


Figure 1: (Left) Stress equals Y times strain. (Right) Stress equals S times shear.

2 Springs and Restoring Forces

- Suppose three springs with equal k constants are connected *in series* (back to back). (a) Show that the effective spring constant is $k_{\text{tot}} = k/3$. (b) Suppose $k = 50 \text{ N m}^{-1}$. If the springs have original length 3 cm, what is the total length if a 1 kg painting is hung from them?

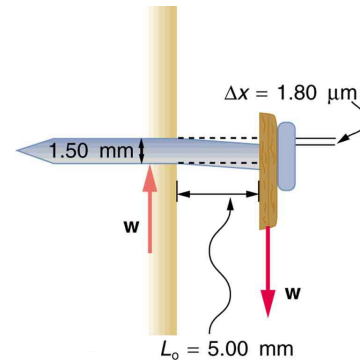


Figure 2: A picture hangs on a nail, causing a shear force.

- Find the mass of the picture hanging from a steel nail, given that the nail bends only 1.80 microns (10^{-6} m). The shear modulus is $80 \times 10^9 \text{ N m}^{-2}$.
- Calculate the change in length of the upper leg bone (the femur) when a 70.0 kg man supports 62.0 kg of his mass on it, assuming the bone to be equivalent to a uniform rod that is 40.0 cm long and 2.00 cm in radius. Young's Modulus for bone is $9 \times 10^9 \text{ N m}^{-2}$.